Glossary

IT Support



Terms and definitions from Course 2

A

A record: The most common resource record, used to point a certain domain name at a certain IPv4 IP address

ACK flag: One of the TCP control flags. ACK is short for acknowledge. A value of one in this field means that the acknowledgment number field should be examined

Acknowledgement number: The number of the next expected segment in a TCP sequence

Ad-Hoc network: A network configuration without supporting network infrastructure. Every device involved with the ad-hoc network communicates with every other device within range, and all nodes help pass along messages

Address class system: A system which defines how the global IP address space is split up

Address Resolution Protocol (ARP): A protocol used to discover the hardware address of a node with a certain IP address

Anycast: A technique that's used to route traffic to different destinations depending on factors like location, congestion, or link health

Application layer: The layer that allows network applications to communicate in a way they understand

Application layer payload: The entire contents of whatever data applications want to send to each other

ARP table: A list of IP addresses and the MAC addresses associated with them

ASN: Autonomous System Number is a number assigned to an individual autonomous system

Asymmetric Digital Subscriber Line (ADSL): A device that establishes data connections across phone lines and different speeds for uploading and downloading data

Automatic allocation: A range of IP addresses is set aside for assignment purposes

B

Baud rate: A measurement of how many bits could be passed across a phone line in a second

Bit: The smallest representation of data that a computer can understand

Bluetooth: The most common short range wireless network

Border Gateway Protocol (BGP): A protocol by which routers share data with each other

Broadband: Any connectivity technology that isn't dial-up Internet

Broadcast: A type of Ethernet transmission, sent to every single device on a LAN

Broadcast address: A special destination used by an Ethernet broadcast composed by all Fs

C

Cable categories: Groups of cables that are made with the same material. Most network cables used today can be split into two categories, copper and fiber

Cable modem termination system: Connects lots of different cable connections to an ISP's core network

Cable modem: A device that sits at the edge of a consumer's network and connects it to the cable modem termination system

Cables: Insulated wires that connect different devices to each other allowing data to be transmitted over them

Caching and recursive name servers: They are generally provided by an ISP or your local network, and their purpose is to store domain name lookups for a certain amount of time

Carrier-Sense Multiple Access with Collision Detection (CSMA/CD): CSMA/CD is used to determine when the communications channels are clear and when the device is free to transmit data

Channels: Individual, smaller sections of the overall frequency band used by a wireless network

Client: A device that receives data from a server

CLOSE: A connection state that indicates that the connection has been fully terminated, and that no further communication is possible

CLOSE_WAIT: A connection state that indicates that the connection has been closed at the TCP layer, but that the application that opened the socket hasn't released its hold on the socket yet

Cloud computing: The concept and technological approach of accessing data, using applications, storing files, etc. from anywhere in the world as long as you have an internet connection

CNAME: A resource record used to map one domain to another

Collision domain: A network segment where only one device can communicate at a time

Computer networking: The full scope of how computers communicate with each other

Connection-oriented protocol: A data-transmission protocol that establishes a connection at the transport layer, and uses this to ensure that all data has been properly transmitted

Connectionless protocol: A data-transmission protocol that allows data to be exchanged without an established connection at the transport layer. The most common of these is known as UDP, or User Datagram Protocol

Copper cable categories : These categories have different physical characteristics like the number of twists in the pair of copper wires. These are defined as names like category (or cat) 5, 5e, or 6, and how quickly data can be sent across them and how resistant they are to outside interference are all related to the way the twisted pairs inside are arranged

Crosstalk: Crosstalk is when an electrical pulse on one wire is accidentally detected on another wire

Cyclical Redundancy Check (CRC): A mathematical transformation that uses polynomial division to create a number that represents a larger set of data. It is an important concept for data integrity and is used all over computing, not just network transmissions

D

Datalink layer: The layer in which the first protocols are introduced. This layer is responsible for defining a common way of interpreting signals, so network devices can communicate

Data offset field: The number of the next expected segment in a TCP packet/datagram

Data packet: An all-encompassing term that represents any single set of binary data being sent across a network link

Data payload section: Has all of the data of the protocols further up the stack of a frame

Demarcate: To set the boundaries of something

Demarcation point: Where one network or system ends and another one begins

Demultiplexing: Taking traffic that's all aimed at the same node and delivering it to the proper receiving service

Destination MAC address: The hardware address of the intended recipient that immediately follows the start frame delimiter

Destination network: The column in a routing table that contains a row for each network that the router knows about

Destination port: The port of the service the TCP packet is intended for

DHCP: A technology that assigns an IP address automatically to a new device. It is an application layer protocol that automates the configuration process of hosts on a network

DHCP discovery: The process by which a client configured to use DHCP attempts to get network configuration information

Dial-up: Uses POTS for data transfer, and gets its name because the connection is established by actually dialing a phone number

DNS zones: A portion of space in the Domain Name System (DNS) that is controlled by an authoritative name server

Domain: Used to demarcate where control moves from a top-level domain name server to an authoritative name server

Domain name: A website name; the part of the URL following www.

Domain Name System (DNS): A global and highly distributed network service that resolves strings of letters, such as a website name, into an IP address

Dotted decimal notation: A format of using dots to separate numbers in a string, such as in an IP address

DSL: Digital subscriber line was able to send much more data across the wire than traditional dial-up technologies by operating at a frequency range that didn't interfere with normal phone calls

DSLAM: Digital Subscriber Line Access Multiplexers are devices that connect multiple DSL connections to a high-speed digital communications channel

Duplex communication: A form of communication where information can flow in both directions across a cable

Duration field: Specifies how long the total frame is

Dynamic allocation: A range of IP addresses is set aside for client devices and one of these IPs is issued to these devices when they request one

Dynamic IP address: An IP address assigned automatically to a new device through a technology known as Dynamic Host Configuration Protocol

E

Error detection: The ability for a protocol or program to determine that something went wrong

Error recovery: The ability for a protocol or program to attempt to fix an error

ESTABLISHED: Status indicating that the TCP connection is in working order, and both sides are free to send each other data

Ethernet: The protocol most widely used to send data across individual links

Ethernet frame: A highly structured collection of information presented in a specific order

EtherType field: It follows the Source MAC Address in a dataframe. It's 16 bits long and used to describe the protocol of the contents of the frame

Exterior gateway: Protocols that are used for the exchange of information between independent autonomous systems

F

Fiber cable: Fiber optic cables contain individual optical fibers which are tiny tubes made of glass about the width of a human hair. Unlike copper, which uses electrical voltages, fiber cables use pulses of light to represent the ones and zeros of the underlying data

FIN: One of the TCP control flags. FIN is short for finish. When this flag is set to one, it means the transmitting computer doesn't have any more data to send and the connection can be closed

FIN_WAIT: A TCP socket state indicating that a FIN has been sent, but the corresponding ACK from the other end hasn't been received yet

Firewall: It is a device that blocks or allows traffic based on established rules

Five layer model: A model used to explain how network devices communicate. This model has five layers that stack on top of each other: Physical, Data Link, Network, Transport, and Application

Fixed allocation: Requires a manually specified list of MAC address and the corresponding IPs

Flag field: It is used to indicate if a datagram is allowed to be fragmented, or to indicate that the datagram has already been fragmented

Flat file: A collection of records/information that follow a consistent format with rules around stored values. On a host computer, one use is to have a list of network address and host name pairs (a hosts file)

Flow label field: 20-bit field that's used in conjunction with the traffic class field for routers to make decisions about the quality of service level for a specific datagram

Fragmentation: The process of taking a single IP datagram and splitting it up into several smaller datagrams

Fragmentation offset field: It contains values used by the receiving end to take all the parts of a fragmented packet and put them back together in the correct order

Frame check sequence: It is a 4-byte or 32-bit number that represents a checksum value for the entire frame

Frame control field: 16 bits long, it contains a number of sub-fields that are used to describe how the frame itself should be processed

Frequency band: A certain section of the radio spectrum that's been agreed upon to be used for certain communications

FTP: An older method used for transferring files from one computer to another, but you still see it in use today

FTTB: Fiber to the building, fiber to the business or even fiber to the basement, since this is generally where cables to buildings physically enter. FTTB is a setup where fiber technologies are used for data delivery to an individual building

FTTH: Fiber to the home. This is used in instances where fiber is actually run to each individual residents in a neighborhood or apartment building

FTTN: Fiber to the neighborhood. This means that fiber technologies are used to deliver data to a single physical cabinet that serves a certain amount of the population

FTTP: Fiber to the premises. FTTH and FTTB may both also be referred to as FTTP

FTTX: Stands for fiber to the X, where the X can be one of many things

Full duplex: The capacity of devices on either side of a networking link to communicate with each other at the exact same time

Fully qualified domain name: When you combine all the parts of a domain together

H

Half-duplex: It means that, while communication is possible in each direction, only one device can be communicating at a time

Handshake: A way for two devices to ensure that they're speaking the same protocol and will be able to understand each other

HDSL: High Bit-rate Digital Subscriber Lines. These are DSL technologies that provision speeds above 1.544 megabits per second

Header checksum field: A checksum of the contents of the entire IP datagram header

Header length field: A four bit field that declares how long the entire header is. It is almost always 20 bytes in length when dealing with IPv4

Hexadecimal: A way to represent numbers using a numerical base of 16

Hop limit field: An 8-bit field that's identical in purpose to the TTL field in an IPv4 header

Host file: It is a flat file that contains, on each line, a network address followed by the host name it can be referred to as

Hub: It is a physical layer device that broadcasts data to everything computer connected to it

Hybrid cloud: Used to describe situations where companies might run things like their most sensitive proprietary technologies on a private cloud or on premise while entrusting their less sensitive servers to a public cloud

Hypervisor: A piece of software that runs and manages virtual machines while also offering guests a virtual operating platform that's indistinguishable from actual hardware

I

IANA: The Internet Assigned Numbers Authority, is a non-profit organization that helps manage things like IP address allocation

ICMP: Internet control message protocol is used by router or remote hosts to communicate error messages when network problems prevent delivery of IP packets

ICMP payload: Piece of the packet which lets the recipient of the message knows which of their transmissions caused the error being reported

Identification field: It is a 16-bit number that's used to group messages together

Infrastructure as a Service (IaaS): A subset of cloud computing where a network and servers are provided for customers to run their services

Instantiation: The actual implementation of something defined elsewhere

Interface: For a router, the port where a router connects to a network. A router gives and receives data through its interfaces. These are also used as part of the routing table

Interior gateway: Interior gateway protocols are used by routers to share information within a single autonomous system

Internet Protocol (IP): The most common protocol used in the network layer

Internet Service Provider (ISP): A company that provides a consumer an internet connection

Internetwork: A collection of networks connected together through routers - the most famous of these being the Internet

IP datagram: a highly structured series of fields that are strictly defined

IP masquerading: The NAT obscures the sender's IP address from the receiver

IP options field: An optional field and is used to set special characteristics for datagrams primarily used for testing purposes

IPv6 tunnel: IPv6 tunnel servers on either end of a connection take incoming IPv6 traffic and encapsulate it within traditional IPv4 datagrams

IPv6 tunnel brokers: Companies that provide IPv6 tunneling endpoints for you, so you don't have to introduce additional equipment to your network

L

Line coding: Modulation used for computer networks

Link-local unicast address: Allow for local network segment communications and are configured based upon a host's MAC address

Listen: It means that a TCP socket is ready and listening for incoming connections

Local Area Network (LAN): A single network in which multiple devices are connected

Loopback address: An IP address that always points to itself. This type of address is used to test internal pathing through the TCP/IP protocols

M

MAC(Media Access Control) address: A globally unique identifier attached to an individual network interface. It's a 48-bit number normally represented by six groupings of two hexadecimal numbers

MAC filtering: Access points are configured to only allow for connections from a specific set of MAC addresses belonging to devices you trust

Mesh networks: Like ad-hoc networks, lots of devices communicate with each other device, forming a mesh if you were to draw lines for all the links between all the nodes

Metered connection: An internet connection where all data transfer usage is tracked. Cell phone plans that have a limit on data usage per month or that charge based on usage are examples of metered connections

Modulation: A way of varying the voltage of a constant electrical charge moving across a standard copper network cable

Multicast: A way of addressing groups of hosts all at once

Multicast frame: If the least significant bit in the first octet of a destination address is set to one, it means you're dealing with a multicast frame. A multicast frame is similarly set to all devices on the local network signal, and it will be accepted or discarded by each device depending on criteria aside from their own hardware MAC address

Multiplexing: It means that nodes on the network have the ability to direct traffic toward many different receiving services

MX record: It stands for mail exchange and this resource record is used in order to deliver email to the correct server

N

Name resolution: This process of using DNS to turn a domain name into an IP address

Network Address Translation (NAT): A mitigation tool that lets organizations use one public IP address and many private IP addresses within the network

Network layer: It's the layer that allows different networks to communicate with each other through devices known as routers. It is responsible for getting data delivered across a collection of networks

Network port: The physical connector to be able to connect a device to the network. This may be attached directly to a device on a computer network, or could also be located on a wall or on a patch panel

Network switch: It is a level 2 or data link device that can connect to many devices so they can communicate. It can inspect the contents of the Ethernet protocol data being sent around the

network, determine which system the data is intended for and then only send that data to that one system

Next header field: Defines what kind of header is immediately after this current one

Next hop: The IP address of the next router that should receive data intended for the destination networking question or this could just state the network is directly connected and that there aren't any additional hops needed. Defined as part of the routing table

Node: Any device connected to a network. On most networks, each node will typically act as a server or a client

Non-metered connection: A connection where your data usage is not tracked or limited, instead you are charged a flat fee for unlimited and unrestricted usage. A Wi-Fi connection is an example of a non-metered connection

Non-routable address space: They are ranges of IPs set aside for use by anyone that cannot be routed to

NS record: It indicates other name servers that may also be responsible for a particular zone

NTP servers: Used to keep all computers on a network synchronized in time

O

Octet: Any number that can be represented by 8 bits

Optical Network Terminator: Converts data from protocols the fiber network can understand to those that are more traditional twisted pair copper networks can understand

Options field: It is sometimes used for more complicated flow control protocols

Organizationally Unique Identifier (OUI): The first three octets of a MAC address

OSI model: A model used to define how network devices communicate. This model has seven layers that stack on top of each other: Physical, Data Link, Network, Transport, Session, Presentation, and Application

P

Padding field: A series of zeros used to ensure the header is the correct total size

Pairing: When a wireless peripheral connects to a mobile device, and the two devices exchange information, sometimes including a PIN or password, so that they can remember each other

Patch panel: A device containing many physical network ports

Payload: The actual data being transported, which is everything that isn't a header

Payload length field: 16-bit field that defines how long the data payload section of the datagram is

Physical layer: It represents the physical devices that interconnect computers

Platform as a service: A subset of cloud computing where a platform is provided for customers to run their services

Point-To-Point VPN: Establishes a VPN tunnel between two sites but VPN tunneling logic is handled by network devices at either side, so that users don't all have to establish their own connections

Pointer resource record: It resolves an IP to a name

Port: It is a 16-bit number that's used to direct traffic to specific services running on a networked computer

Port forwarding: A technique where specific destination ports can be configured to always be delivered to specific nodes

Port preservation: A technique where the source port chosen by a client, is the same port used by the router

Preamble: The first part of an Ethernet frame, it is 8 bytes or 64 bits long and can itself be split into two sections

Presentation layer: It is responsible for making sure that the unencapsulated application layer data is actually able to be understood by the application in question

Private cloud: When a company owns the services and the rest of the cloud infrastructure, whether on-site or in a remote data center

Protocol: A defined set of standards that computers must follow in order to communicate properly is called a protocol

Protocol field: A protocol field is an 8-bit field that contains data about what transport layer protocol is being used

Proxy service: A server that acts on behalf of a client in order to access another service

PSH flag: One of the TCP control flags. PSH is short for push. This flag means that the transmitting device wants the receiving device to push currently- buffered data to the application on the receiving end as soon as possible

Public cloud: The cloud services provided by a third party

Public DNS servers: Name servers specifically set up so that anyone can use them for free

Q

Quad A (AAAA) record: It is very similar to an A record except that it returns in IPv6 address instead of an IPv4 address

R

Receiving address: The MAC address of the access point that should receive the frame

Recursive name servers: Servers that perform full DNS resolution requests

Registrar: An organization responsible for assigning individual domain names to other organizations or individuals

Reverse lookup zone files: They let DNS resolvers ask for an IP, and get the FQDN associated with it returned

Reverse proxy: A service that might appear to be a single server to external clients, but actually represents many servers living behind it

Round robin: It is a concept that involves iterating over a list of items one by one in an orderly fashion

Router: A device that knows how to forward data between independent networks

Routing protocols: Special protocols the routers use to speak to each other in order to share what information they might have

RST flag: One of the TCP control flags. RST is short for reset. This flag means that one of the sides in a TCP connection hasn't been able to properly recover from a series of missing or malformed segments

S

Sequence control field: A field that is 16 bits long and mainly contains a sequence number used to keep track of ordering the frames

Sequence number: A 32-bit number that's used to keep track of where in a sequence of TCP segments this one is expected to be

Server: A device that provides data to another device that is requesting that data, also known as a client

Server or Service: A program running on a computer waiting to be asked for data

Service type field: A eight bit field that can be used to specify details about quality of service or QoS technologies

Session layer: The network layer responsible for facilitating the communication between actual applications and the transport layer

Short-range wireless network: It is what mobile devices uses to connect to their peripherals

Simplex communication: A form of data communication that only goes in one direction across a cable

Socket: The instantiation of an endpoint in a potential TCP connection

Software as a Service (SaaS): A way of licensing the use of software to others while keeping that software centrally hosted and managed

Source MAC address: The hardware address of the device that sent the ethernet frame or data packet. In the data packet it follows the destination MAC address

Source port: A high numbered port chosen from a special section of ports known as ephemeral ports

SRV record: A service record used to define the location of various specific services

Start Frame Delimiter (SFD): The last byte in the preamble, that signals to a receiving device that the preamble is over and that the actual frame contents will now follow

Start of authority: A declaration of the zone and the name of the name server that is authoritative for it

Static IP address: An IP address that must be manually configured on a node

Subnet mask: 32-bit numbers that are normally written as four octets of decimal numbers

Subnetting: The process of taking a large network and splitting it up into many individual smaller sub networks or subnets

Symmetric Digital Subscriber Line (SDSL): A device that establishes data connections across phone lines and has upload and download speeds that are the same

SYN flag: One of the TCP flags. SYN stands for synchronize. This flag is used when first establishing a TCP connection and make sure the receiving end knows to examine the sequence number field

SYN_RECEIVED: A TCP socket state that means that a socket previously in a listener state, has received a synchronization request and sent a SYN_ACK back

SYN_SENT: A TCP socket state that means that a synchronization request has been sent, but the connection hasn't been established yet

T

T-Carrier technologies: Technologies Invented to transmit multiple phone calls over a single link. Eventually, they also became common transmission systems to transfer data much faster than any dial-up connection could handle

TCP checksum: A mechanism that makes sure that no data is lost or corrupted during a transfer

TCP segment: A payload section of an IP datagram made up of a TCP header and a data section

TCP window: The range of sequence numbers that might be sent before an acknowledgement is required

Time-To-Live field (TTL): An 8-bit field that indicates how many router hops a datagram can traverse before it's thrown away

Top Level Domain (TLD): The top level of the DNS or the last part of a domain name. For example, the “com” in www.weather.com

Total hops: The total number of devices data passes through to get from its source to its destination. Routers try to choose the shortest path, so fewest hops possible. The routing table is used to keep track of this

Total length field: A 16-bit field that indicates the total length of the IP datagram it's attached to

Traffic class field: An 8-bit field that defines the type of traffic contained within the IP datagram and allows for different classes of traffic to receive different priorities

Transmission Control Protocol (TCP): The data transfer protocol most commonly used in the fourth layer. This protocol requires an established connection between the client and server

Transmitter address: The MAC address of whatever has just transmitted the frame

Transport layer: The network layer that sorts out which client and server programs are supposed to get the data

TTL: The lifetime limit of data given in seconds. This number can be configured by the owner of a domain name for how long a name server is allowed to cache in entry before it should discard it and perform a full resolution again

Twisted pair cable: The most common type of cabling used for connecting computing devices. It features pairs of copper wires that are twisted together

Two-factor authentication: A technique where more than just a username and password are required to authenticate. Usually, a short-lived numerical token is generated by the user through a specialized piece of hardware or software

TXT record: It stands for text and was originally intended to be used only for associating some descriptive text with a domain name for human consumption

Types of DNS servers: There are five primary types of DNS servers; caching name servers, recursive name servers, root name servers, TLD name servers, and authoritative name servers

U

Unicast transmission: A unicast transmission is always meant for just one receiving address

Urgent pointer field: A field used in conjunction with one of the TCP control flags to point out particular segments that might be more important than others

URG flag: One of the TCP control flags. URG is short for urgent. A value of one here indicates that the segment is considered urgent and that the urgent pointer field has more data about this

User Datagram Protocol (UDP): A transfer protocol that does not rely on connections. This protocol does not support the concept of an acknowledgement. With UDP, you just set a destination port and send the data packet

V

Version field: First field in an IP header that specifies the version of IP

Virtual LAN (VLAN): It is a technique that lets you have multiple logical LANs operating on the same physical equipment

Virtual Private Network (VPN): A technology that allows for the extension of a private or local network, to a host that might not work on that same local network

Virtualization: A single physical machine called a host runs many individual virtual instances called guests

VLAN header: A piece of data that indicates what the frame itself is. In a data packet it is followed by the EtherType

W

Wide area network: Acts like a single network but spans across multiple physical locations. WAN technologies usually require that you contract a link across the Internet with your ISP

Wi-Fi Protected Access (WPA): A security program that uses a 128-bit key to protect wireless computer networks, which makes it more difficult to crack than WEP

Wired Equivalence Privacy (WEP): An encryption technology that provides a very low level of privacy. WEP should really only be seen as being as safe as sending unencrypted data over a wired connection

Wireless access point: A device that bridges the wireless and wired portions of a network

Wireless LANS (WLANS): One or more access points act as a bridge between a wireless and a wired network

Wireless networking: Networks you connect to through radios and antennas

Z

Zone Files: Simple configuration files that declare all resource records for a particular zone